

E - WOA : An Enhanced Whale Optimization Algorithm for Energy Optimization in Wireless Sensor Networks

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Abstract—Wireless Sensor Networks play a role in modern applications like environmental monitoring, healthcare and smart infrastructure. The problem is that sensor nodes have limited battery capacity which affects how long the network lasts and how well it performs. To make Wireless Sensor Networks last longer we need to use energy. In this work we propose an Enhanced Whale Optimization Algorithm for energy optimization in Wireless Sensor Networks. The Enhanced Whale Optimization Algorithm improves the Whale Optimization Algorithm by using adaptive control parameters, a distance-based weighting mechanism and a Gaussian-based local search strategy. These improvements help the algorithm balance exploration and exploitation leading to convergence and energy-efficient solutions. We test the Enhanced Whale Optimization Algorithm through simulation in a two- sensor field and compare it with other optimization techniques like Genetic Algorithm, Differential Evolution and the standard Whale Optimization Algorithm. The evaluation of this algorithm is based on things like energy consumption how long the Wireless Sensor Networks last and how quickly they converge. Simulation results show that the Enhanced Whale Optimization Algorithm uses energy and makes Wireless Sensor Networks last longer than other methods. It turns out that the Enhanced Whale Optimization Algorithm really helps with energy optimization in Wireless Sensor Networks that do not have resources. The Enhanced Whale Optimization Algorithm reduces energy consumption. Extends Wireless Sensor Networks lifetime. The findings also show that the Enhanced Whale Optimization Algorithm provides a solution for energy optimization in Wireless Sensor Networks. This algorithm is good, for Wireless Sensor Networks. The findings confirm that the Enhanced Whale Optimization Algorithm provides a solution for energy optimization in Wireless Sensor Networks with limited resources. The results also show that the Enhanced Whale Optimization Algorithm is scalable and can be used in Wireless

Sensor Networks applications. The proposed approach helps to make Wireless Sensor Networks more efficient and long-lasting. The Enhanced Whale Optimization Algorithm achieves this by optimizing energy consumption, in Wireless Sensor Networks. This leads to performance and longer lifetime of Wireless Sensor Networks.

Index Terms—Wireless Sensor Networks, Energy Optimization, Whale Optimization Algorithm, Enhanced WOA, Meta-heuristic Optimization, Network Lifetime, Energy Efficiency, Swarm Intelligence.

I. INTRODUCTION

Wireless Sensor Networks are really helpful for monitoring and collecting data in areas, such as environmental monitoring, healthcare, industrial automation and smart infrastructure. A typical Wireless Sensor Network has sensor nodes that can sense, process and send data to a central base station or sink. These networks have become very important in intelligent systems because they are easy to set up and can be used in many different situations. One of the big challenges with Wireless Sensor Networks is that the sensor nodes have limited energy. These nodes are usually powered by batteries. They have to be placed in far away locations or in spots that are very hard to reach. This makes it very difficult to replace the batteries of these nodes or charge these nodes up again. These nodes have to keep working for a time because of this situation, with these nodes. So energy efficiency becomes a design consideration for making the network last longer and keeping communication reliable. Communication operations, data transmission use a lot of energy in Wireless Sensor